Artificial Economies for Learning Agents: Testing Whether Institutions Stay Robust Across Learning Architectures

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Abstract

This paper introduces a code-first experimental platform for studying whether economic mechanisms remain robust when the strategic agents inside them are adaptive learners rather than equilibrium solvers. The platform implements a shared `World/Agent/Institution` interface, repeated multiseed evaluation, exploitability tests, and interchangeable mind classes ranging from random and tabular Q-learning agents to PyTorch DQN, PPO, decorrelated independent-DQN, and centralized-critic agents. The first validated environment is a repeated two-firm pricing arena with six regulatory mechanisms. In this world, price caps robustly reduce exploitability across mind classes, but their effect on collusion is metric- and architecture-dependent. A new paired-seed audit shows that DQN's positive profit-under-cap result is not visible in 1000-step traces; it emerges with training budget, becomes mostly positive by 10,000 steps, and is positive in the 40,000-step full table. The second environment, Resource Island, extends the same interface to a spatial inventory economy and reveals a different lesson: institution conclusions are only meaningful when the world actually exercises the institutional channel. The reported Resource Island configuration therefore uses contested resources, unequal trades, specialization pressure, and activation diagnostics before interpreting property rights, trade controls, and reputation. Three additional environments extend the suite: Auction House provides a benchmarked auction-design testbed, Public Goods exposes common-pool extraction and contribution incentives, and Labor Market tests learned reporting under deferred acceptance. The current contribution is both empirical and methodological: it reports initial evidence that institutional robustness changes across learning architectures, and it documents the validation discipline needed to distinguish real economic findings from inactive-mechanism or short-horizon artifacts.

1 Introduction

Economic mechanisms are usually evaluated against a model of strategic behavior: equilibrium prices, truthful bidding, efficient allocation, or some other theoretical benchmark. That evaluation is only as good as the behavioral model behind it. This paper asks what happens when institutions designed around classical economic assumptions are placed in front of learning agents: which guarantees survive, which break, and does the break depend on the learning architecture?

We answer this with five small, controlled economic worlds, presented below in order of benchmark strength: a sealed-bid auction (Auction House), a two-sided labor match (Labor Market), a pricing duopoly (Pricing Arena), a common-pool resource game (Public Goods), and a scarce-resource trade economy (Resource Island). Each is paired with a classical benchmark and a shared ladder of learning architectures: random behavior, tabular Q-learning, deep Q-networks, proximal policy optimization, independent multi-agent Q-learning, and centralized-critic training. These architectures are not a scale from weaker to stronger. The question at stake in each world is not which architecture performs best, but which of the institution's guarantees hold up as the behavioral assumptions behind the benchmark are changed.

Across the five worlds, the same failure modes recur in different forms: an institution can look effective on one metric while a different strategic channel stays open; an apparent effect can depend on training horizon and reverse sign with more training budget; a learning architecture can fail to recover a benchmark that classical theory says is optimal; and an institution can raise a headline welfare number while quietly eroding the property, stability, sustainability, truthfulness, that the institution actually exists to protect. The next section sets out, world by world, the classical benchmark each environment owes and what claim boundary follows from it, while the world sections report where each of these failure modes appears, and where the evidence is instead null or still provisional.

2 Related Work

This paper connects five economic literatures to a shared reinforcement-learning testbed. Each world below is built to test one specific classical prediction. The Pricing Arena is motivated by work on algorithmic collusion: repeated-game Q-learning agents can learn to sustain prices above the competitive level, without any explicit communication between them [Calvano et al., 2020, 2021]. The exploitability check used in this paper follows the same concern one step further. A policy's average performance across a run does not show whether that policy is robust to a deviating opponent, so this paper measures both. Auction House is based on classical auction theory. In a second-price auction, bidding one's true value is a dominant strategy, meaning it is optimal regardless of what other bidders do [Vickrey, 1961]. In a first-price auction, by contrast, the optimal strategy is to bid below one's true value (shading), and reserve prices create a known tradeoff between the seller's expected revenue and the efficiency of the final allocation [Myerson, 1981, Milgrom, 2004].

Public Goods and Resource Island are both based on common-pool resource and public-goods theory. The standard prediction in this literature is that, absent some form of monitoring, reputation, or locally appropriate governance, a shared resource will be overused or a public good will be underfunded, because each agent's private incentive does not account for the cost imposed on everyone else [Samuelson, 1954, Ostrom, 1990]. Labor Market is based on two-sided matching theory. Under the worker-proposing deferred acceptance algorithm, the resulting match is guaranteed to be stable (no worker and employer who are not matched to each other would both prefer to be), and truthful reporting of preferences is guaranteed to be optimal for the proposing side [Gale and Shapley, 1962, Roth and Sotomayor, 1990, Roth, 1984]. These are proven properties of the mechanism, not empirical tendencies, which is what makes them useful as benchmarks.

The learner suite is built from five standard reinforcement learning methods: tabular Q-learning, deep Q-networks (DQN) [Sutton and Barto, 2018, Mnih et al., 2015], proximal policy optimization (PPO) [Schulman et al., 2017], independent Q-learning [Tan, 1993], and centralized-critic training [Lowe et al., 2017]. What is different here is that the suite is used to test assumptions, not to compare performance. For example, second-price truthfulness and deferred-acceptance strategy-proofness are proofs about a single, fixed learner responding to a fixed mechanism; independent Q-learning and centralized-critic training instead put multiple agents in motion at the same time, which is not the setting those proofs were written for.

3 Theory Obligations

Every world in this paper owes something to the theory it is built on: a specific prediction it must be checked against, a specific set of measurements that follow from that prediction, and a specific boundary on what the resulting evidence can support. This section states that obligation for each world before any learned-agent result is discussed. The obligation exists because passing tests and producing a complete set of output files is not the same thing as having tested the institution a world claims to test.

The five worlds do not owe the same kind of proof. Grouping them by the strength of their classical prediction gives three tiers: proven mechanism properties, where a learned agent's behavior can be checked directly against a known correct answer; bracketed or reference-point theory, where exact values exist but no single predicted outcome for the object under test; and activation, where no external benchmark exists at all and the obligation is instead to show the institution was ever placed under real economic pressure.

Auction House and Labor Market belong to the first tier. In Auction House, the classical predictions are exact: truthful bidding in a second-price auction, bid shading in a first-price auction, and a known revenue-efficiency tradeoff under a reserve price. Because the correct behavior is known, this world measures distance from it directly, as revenue, allocative efficiency, bidder surplus, ex-post regret, over- and underbidding, shading distance, and the shape of the learned bid curve against the truthful or shaded reference, rather than raw payoff alone. The resulting claim is bounded to benchmark deviation under different learners; convergence to auction equilibrium is not asserted. In Labor Market, a lower truthful-report rate under some learner is not by itself evidence of profitable manipulation, given the proposing side's strategy-proofness guarantee. Match rate, blocking-pair stability, truthful-report rate, and welfare are therefore reported together, since a learner can trade one against another, and any claim about manipulation is bounded to the worker-proposing, canonical version of the mechanism implemented here.

Pricing Arena and Public Goods belong to the second tier. Pricing Arena has two exact static prices, the Nash price and the joint-profit price, but no closed-form prediction for behavior in the repeated game itself, only a range bounded by the two. A collusion claim in this world is therefore only as credible as the decomposition behind it: price, profit, quantity, welfare, and exploitability are reported together and checked against each other, and the resulting claim is limited to finite-run, regulatory evidence in a stylized duopoly, not a theorem or a real-market antitrust result. Public Goods is bracketed rather than pointed: theory predicts underprovision when private incentives diverge from the social good, with free-riding as a lower bound and the social optimum as an upper one, and nothing more specific in between. The obligation this creates is to keep the welfare number honest against the underlying resource state, since an institution can raise measured welfare through reward accounting without changing the incentive to extract or contribute at all. Contribution, extraction, sustainability, inequality, and tax revenue are therefore reported as separate quantities from welfare, so that an accounting effect is not mistaken for a real one.

Resource Island belongs to the third tier alone. It carries no external closed-form benchmark, so its obligation is not how close a learned agent gets to a known correct value, but whether the institution being measured was ever placed under real economic pressure in the first place. A property-rights rule is not being tested if no non-owner ever has reason to access a claimed resource; a trade price control is not being tested if every trade happens to be one-for-one already. Survival, sustainability, trade counts, property pressure, specialization, and institution-block counters exist to answer that prior question before any welfare number from this world is trusted. This is not a weaker test than the other four worlds run; it is a test of a condition the other four simply assume already holds, and the Resource Island section reports institution effects only after those activation conditions are measured.

4 Platform

4.1 Interfaces

The codebase is organized around three abstractions:

- A `World` defines state transitions, observations, metrics, and rendering state.
- An `Agent` or mind maps observations to actions and updates from rewards and next observations.
- An `Institution` modifies rules, rewards, constraints, or information inside a world.

This separation is not cosmetic. It makes it possible to ask whether a result is about a world, an institution, or a learner. It also creates a regression test: if a new world requires rewriting the shared learning interface, then the platform abstraction has failed.

4.2 Validation Discipline

The platform treats a result as reportable only after four checks. First, the mechanics must be unit-tested: allocation, movement, payment, trade, collision, or reward rules must match hand-computed cases. Second, where theory provides a known answer, the implementation must recover or bracket that answer before learned-policy output is interpreted. The current known-answer runner reports 19/19 passing checks across Pricing Arena N-firm static Nash references, Auction House truthfulness and bid-shading cases, Public Goods free-rider and cooperative brackets, Labor Market deferred-acceptance benchmark cases, and Resource Island gather bounds. Third, the experiment must be replicated over seeds, with per-seed and aggregate CSV outputs. Fourth, the institution must actually activate. An institution that never faces the behavior it is meant to govern produces no evidence about that institution, only about the world design. The Resource Island section gives a concrete case. The second new validation layer is mechanism tracing. The trace runner produces per-step rows and algebraic decomposition rows for all five worlds and all thesis-facing minds. These traces do not replace full-run evidence; they show the channel behind a claim and can expose contradictions between short traces and full-run results. The Pricing Arena section gives a concrete case.

5 Auction House

5.1 World

Auction House is a single-item auction environment with independent private values, discrete bid actions, deterministic tie-breaking, and configurable payment rules. Each bidder observes its own valuation bin and chooses a bid from a fixed grid. The implemented auction formats are second-price, first-price, and a simplified ascending-clock abstraction; the second-price format also supports a reserve-price variant. Two information variants modify what a bidder observes without changing the allocation or payment rule: a public-signal version mixes a bidder's own value with rival-value information, and a noisy-signal version perturbs the observed value bin directly.

The world records seller revenue, bidder surplus, realized welfare, maximum feasible welfare, allocative efficiency, ex-post regret, overbidding, underbidding, truthful-bid distance, first-price shading distance, and no-sale rate.

5.2 Known-Answer Validation

The benchmark obligations for Auction House are stated in Section 3: truthful second-price bidding, first-price bid shading, and the reserve-price revenue-efficiency tradeoff. The validation layer checks those obligations before learned bidders are interpreted. The known-answer runner confirms that truthful bidding has zero grid regret in second-price auctions for both two and three bidders, and that the first-price shading benchmark bids strictly below value at the same bidder counts. Separately, unit tests cover allocation, first-price and second-price payment, reserve and no-sale behavior, clock clearing, deterministic tie-breaking, and the information-policy observation transforms.

Table 1 lists all six implemented scenarios with their validation status. The three sealed-bid formats carry the most direct benchmarks and the most complete evidence; the clock and information variants are validated and interpretable but are read here as extensions rather than as central results.

Scenario	Validation	Role
Second price	Known-answer checks pass; full Q-learning and full learner-suite runs	Central benchmark
First price	Full Q-learning and full learner-suite runs	Central benchmark
Reserve	Full Q-learning and full learner-suite runs	Central benchmark
Clock	Full learner-suite runs; Q-learning smoke only	Benchmark-compatible extension
Public signal	Full learner-suite runs	Information stress test
Noisy signal	Full learner-suite runs	Information stress test

Table 1: Auction House scenarios and validation status.

5.3 Sealed-Bid Q-Learning Results

Table 2 shows the full $n = 20$ Q-learning run across the three sealed-bid scenarios. The world reproduces the intended economic structure, but learned bidders do not fully recover the theoretical benchmark in any of the three. Second-price allocative efficiency is 0.734 against a truthful benchmark of 1.0. First-price bidding shows the expected shading direction clearly, with underbidding at 0.814 and overbidding at 0.066. The reserve variant raises seller revenue from 3.100 in plain second-price to 3.938, while lowering realized welfare from 6.241 to 6.060 and producing a positive no-sale rate.

Scenario	Revenue	Welfare	Efficiency	Regret	Underbid	No Sale
Second price	3.100	6.241	0.734	0.270	0.423	0.000
First price	3.453	6.318	0.769	1.361	0.814	0.000
Second price + reserve	3.938	6.060	0.796	0.150	0.441	0.172

Table 2: Auction House, Q-learning full run over sealed-bid auction formats.

First price is the most direct recovery of a predicted direction: learned bidders shade below value, as theory predicts, even though they do not converge tightly enough to eliminate regret. Second price gives a different kind of evidence: the mechanism is exactly strategy-proof, but finite learned bidding does not fully reach the dominant strategy, and the shortfall appears as lower allocative efficiency and positive regret rather than as a code or mechanism defect. The reserve result follows the textbook tradeoff direction: revenue rises, welfare falls, and a positive share of auctions produce no sale.

5.4 Cross-Mind Sealed-Bid Results

Table 3 reports second-price results across the full learner suite, where the benchmark is most direct. PPO is closest to the intended outcome of every learner tested: lowest ex-post regret at 0.187, highest welfare at 6.389, and highest allocative efficiency at 0.762. DQN and independent-DQN raise seller revenue above the Q-learning baseline, but with lower efficiency and larger truthful-bid distance; the revenue gain does not come from bidding closer to the truthful benchmark. The centralized critic has the largest benchmark deviation in this table: revenue is close to zero, allocative efficiency falls to 0.522, and regret rises to 1.436.

Mind	Revenue	Welfare	Efficiency	Regret	Truthful Dist.
Q-learning	3.100	6.241	0.734	0.270	1.996
DQN	3.593	6.115	0.705	0.260	2.392
PPO	3.233	6.389	0.762	0.187	1.780
Independent-DQN	3.630	6.124	0.707	0.280	2.294
Centralized critic	0.063	5.061	0.522	1.436	4.577

Table 3: Second-price auction results by learner. Lower regret and truthful-bid distance are better; the benchmark allocative efficiency is 1.0.

Table 4 gives the corresponding first-price results. Q-learning, DQN, PPO, and independent-DQN all show substantial underbidding, between 0.716 and 0.814, so every non-centralized learner recovers the correct qualitative direction, even though none reaches the grid benchmark exactly. Centralized critic again has the largest deviation: lower revenue, lower efficiency, and the highest regret of any learner in either sealed-bid format.

Mind	Revenue	Welfare	Efficiency	Regret	Underbid
Q-learning	3.453	6.318	0.769	1.361	0.814
DQN	3.696	6.409	0.791	1.008	0.723
PPO	3.408	6.365	0.771	0.937	0.716
Independent-DQN	3.412	6.425	0.798	1.048	0.753
Centralized critic	1.280	5.016	0.515	1.974	0.832

Table 4: First-price auction results by learner. Underbidding is expected under first-price bid shading, but regret measures remaining distance from a best response on the bid grid.

The centralized critic's deviation is consistent across both mechanisms and is not an artifact of one auction format. It has the largest benchmark deviations in both second-price and first-price settings, which suggests difficulty with the asymmetric private-information structure rather than with one specific payment rule.

5.5 Reserve, Clock, and Information Variants

The reserve variant reproduces the expected tradeoff across every learner tested, not only Q-learning. DQN and independent-DQN both post reserve revenues above 4.3, roughly a full point higher than their no-reserve second-price revenue, while bidder surplus falls in every case. This is the standard reserve-price direction rather than a new theoretical claim, and it is reported because it holds more uniformly than the second-price and first-price benchmark-recovery results.

The clock and information variants extend the platform beyond the three benchmarked sealed-bid formats. PPO remains closest to the benchmark in the clock abstraction, with welfare 6.402, efficiency 0.769, and regret 0.186, closely tracking its own second-price numbers. The information variants show that changing observations does not automatically improve bidding behavior: for DQN and independent-DQN, public-signal second-price auctions produce lower allocative efficiency than the corresponding private-signal second-price runs, and centralized critic's regret rises from 1.436 in the baseline second-price setting to 1.487 under the public signal and 1.512 under noise. These variants are validated and interpretable, but they are read as stress tests of the auction interface rather than as settled information-design results, since neither has the sealed-bid formats' full Q-learning evidence base.

5.6 Relation to the Main Question

Auction House separates two objects that this paper treats as distinct: the validity of a mechanism and whether learned behavior recovers what the mechanism guarantees. The mechanism layer is exact and validated: second-price truthfulness, first-price shading, and reserve revenue-efficiency behavior all pass known-answer checks before any learned bidder is examined. Learned behavior does not fully match that layer. Even in a clean second-price auction, no learner in the suite converges all the way to truthful, efficient bidding within the tested training budget; first-price learners recover the correct shading direction without reaching a zero-regret benchmark; reserve pricing reproduces its textbook tradeoff consistently across learners.

The gap between mechanism and learned behavior is architecture-dependent. PPO sits closest to the benchmark across second-price, first-price, and clock settings. DQN and independent-DQN typically raise seller revenue while moving farther from truthful bidding, a different kind of deviation than PPO's. The centralized critic has the largest deviations in almost every non-reserve scenario, a pattern specific to this world's private-information structure that does not appear in the same form in Pricing Arena or Resource Island.

Two limits follow from this evidence. First, the result is a benchmark-deviation claim, not a claim that learned bidders converge to auction equilibrium. Second, this platform studies fixed mechanisms with learned bidders, which is different from learned mechanism design in the sense of RegretNet-style work [Dütting et al., 2019], where the auction rule itself is learned. Auction House does not attempt that here: the auction format is fixed, and only the bidding behavior is learned.

6 Labor Market

6.1 World

Labor Market is a two-sided matching world. Workers are learning agents; employers hold fixed, non-learning preferences over workers. In each round, a worker reports a top choice among available employers, and the world constructs a reported preference profile from those actions. A deferred-acceptance procedure then resolves the reported profile into a matching, and each worker's reward is the utility of its matched outcome [Gale and Shapley, 1962, Roth and Sotomayor, 1990]. Unlike the other four worlds, not every economic actor in Labor Market learns: only the workers do, while employers remain fixed preference holders throughout. That asymmetry is part of the test. It checks whether the shared learner suite behaves consistently when applied to only one side of a two-sided mechanism.

The world records match rate, worker welfare, employer welfare, total welfare, blocking-pair counts, a stability measure, truthful-report rate, manipulation-gain diagnostics, and fixed hand-checkable benchmark cases.

6.2 Known-Answer Validation

The benchmark obligations for Labor Market are stability and proposing-side strategy-proofness under worker-proposing deferred acceptance. Four fixed cases validate that mechanism layer before learned results are interpreted: a stable, fully truthful matching case, which must pass with zero blocking pairs; a deliberately forced unstable case, which must be identified by the blocking-pair check; a welfare-accounting case, which must sum payoffs correctly; and a direct check that no worker has a profitable deviation from truthful reporting under standard worker-proposing deferred acceptance. All four pass. This last case matters for interpretation: a later decline in truthful-report rate is not evidence that the implemented mechanism fails to be strategy-proof. It is a behavioral outcome of the learner.

6.3 Baseline Result

Table 5 shows the full $n = 20$ Q-learning run. Match rate is perfect, stability is high, and truthful reporting remains below the exact optimum confirmed by the benchmark case, at 0.786 rather than 1.0. Total welfare also falls below the truthful benchmark welfare of 4.807. The manipulation-gain diagnostic is close to zero, at 0.003, which indicates that lower truthfulness is not the same thing as profitable strategic misreporting in this setup.

Metric	Value
Match rate	1.000
Stability	0.951
Truthful-report rate	0.786
Total welfare	3.639
Benchmark truthful welfare	4.807
Blocking pairs	0.057
Manipulation gain	0.003

Table 5: Labor Market, Q-learning full run.

6.4 Cross-Mind Results

Table 6 reports the same measurements across the full learner suite. Match rate is 1.0 for every learner, so the relevant cross-mind variation is in stability and truthfulness. DQN is the most stable learner in the table, at 0.975, while also reporting truthfully less often than Q-learning. PPO holds stability close to Q-learning's level but reduces truthful reporting to 0.575. Independent-DQN raises welfare slightly above Q-learning while giving up a moderate amount of truthfulness. The centralized critic is the main exception: it produces the highest welfare in the table, 3.821, while stability falls to 0.749 and truthful-report rate falls to 0.433, both below every other learner.

Mind	Match Rate	Stability	Truthful Report	Welfare
Q-learning	1.000	0.951	0.786	3.639
DQN	1.000	0.975	0.704	3.643
PPO	1.000	0.950	0.575	3.643
Independent-DQN	1.000	0.955	0.672	3.658
Centralized critic	1.000	0.749	0.433	3.821

Table 6: Labor Market, full learner suite under deferred acceptance.

These results do not support the claim that worker-proposing deferred acceptance loses strategy-proofness when the reporting side is learned. The manipulation-gain diagnostic remains near zero or negative for every learner in the table, including the centralized critic. A lower truthful-report rate is therefore not evidence that a learner found and exploited a profitable misreport. The appropriate interpretation is behavioral: learners can converge on non-truthful reports that do not pay off in the benchmark sense, while still changing realized stability and welfare through the policies they learn.

6.5 Relation to the Main Question

The centralized-critic row in Table 6 is this world's clearest metric conflict. Centralized critic produces the highest total welfare of any learner tested, 3.821 against Q-learning's 3.639. Read in isolation, that aggregate suggests improvement. Read against the benchmark properties, the same row is weaker: stability falls from 0.951 to 0.749 and truthful reporting falls from 0.786 to 0.433. The result therefore separates payoff improvement from preservation of the mechanism's intended properties.

Labor Market's benchmark is exact, but the exactness attaches to properties of the matching rather than to a scalar payoff. Match rate and stability survive the move to learned worker reports for four of the five learners; truthfulness falls for every neural or MARL learner relative to Q-learning, without that fall constituting evidence of profitable manipulation under the mechanism's own diagnostic. Centralized critic gives the clearest result in this world: it raises welfare furthest above the Q-learning baseline while degrading stability and truthfulness furthest below it.

7 Pricing Arena

7.1 World

Pricing Arena is a repeated two-firm pricing market. Each firm chooses a price from a discrete grid every round. Consumers divide between the two firms according to a noisy logit demand model, so a firm's realized quantity depends on both its own price and its rival's. Profit is price minus cost, times realized quantity, and learning agents update from repeated play against each other. Six mechanisms are implemented on top of this base game: no regulation, a price cap, a tax on high prices, random audits, a direct anti-collusion penalty, and demand shocks.

7.2 Metrics

The world reports average price, total profit, consumer surplus, welfare, price dispersion, exploitability, and two collusion indexes. The first is a price-normalized proxy used since early versions of the project. The second is a profit-normalized index, used because it maps directly onto the collusion measure common in the algorithmic-pricing literature:

$$\mathrm{CI}_{profit} = \frac{\pi_{observed} - \pi_{Nash}}{\pi_{monopoly} - \pi_{Nash}}. \tag{1}$$

Both are reported together, since the two can move in different directions, and that divergence is itself informative about which channel a mechanism is actually affecting.

7.3 Known-Answer Validation

Before any learned result is interpreted, the static Nash best response is checked directly against the world's own demand and cost parameters, for $n = 2$, 3, 4, and 5 firms. All four checks pass. This does not validate anything about learned behavior; it validates that the reference point learned behavior is measured against is not arbitrary.

7.4 Baseline and Main Result

Table 7 shows the unregulated baseline for each learner. Q-learning and the centralized critic sit at the higher-price, higher-collusion end of the range; DQN, PPO, and independent-DQN sit lower, with lower exploitability.

Mind	Avg. Price	Profit	Welfare	Collusion Index	Exploitability
Q-learning	5.487	365.3	810.4	0.541	95.7
DQN	3.611	236.3	832.8	0.202	20.3
PPO	3.681	260.0	839.3	0.215	14.2
Independent-DQN	4.078	269.9	827.9	0.285	26.6
Centralized critic	5.925	288.6	791.6	0.618	147.5

Table 7: Pricing Arena, no-regulation baseline, by learner.

Table 8 shows the change under a price cap, relative to that baseline, for every learner. The exploitability column is the clearest result in this world: it falls for every learner, without exception. The profit column is not: it falls for Q-learning and random play, in line with the conventional expectation that a cap disciplines pricing behavior, but it rises for DQN, independent-DQN, and the centralized critic.

Mind	Price Δ	Profit Δ	Welfare Δ	Exploitability Δ
Q-learning	−0.759	−22.916	+19.567	−55.366
Random	−1.185	−16.028	+34.779	−31.775
DQN	+0.388	+55.358	+6.294	−11.619
PPO	−0.042	−6.558	−1.198	−5.704
Independent-DQN	−0.288	+3.093	+12.183	−14.871
Centralized critic	−1.337	+10.347	+28.668	−98.274

Table 8: Price cap versus no regulation, change by learner, n=20 full run.

The pattern is a guardrail that holds on the axis it was built for and does not hold on an adjacent one. A price cap constrains the price level directly, and every learner's exploitability falls as a result, since a capped price cannot be pushed into the range a deviating opponent could exploit most severely. But the cap does not constrain profit directly, only through price, and for three of the five learners the resulting change in realized quantity and market structure more than offsets the lower price. Q-learning follows the standard price-discipline expectation more closely than DQN, independent-DQN, or the centralized critic.

7.5 The Training-Budget Correction

The DQN result above was not visible in the first version of this analysis. A short, 1000-step mechanism trace, run to check the channel behind the headline number, showed DQN profit falling under the cap for every seed tested, the opposite of Table 8. This discrepancy was resolved by running the same paired-seed comparison at a range of training budgets, from the short trace up to the full 40,000-step run the headline table is drawn from.

Source	Steps	Mean Profit Δ	Positive Seed Share	Mean Quantity Δ
fresh audit	1,000	−11.711	0.10	3.220
fresh audit	5,000	+1.730	0.50	4.102
fresh audit	10,000	+35.448	0.80	3.540
full table	40,000	+55.358	0.85	0.226

Table 9: DQN price-cap profit delta by training budget. Values are paired seed deltas, price cap minus no regulation.

The sign flips between 1,000 and 5,000 steps and is positive with a clear majority of seeds by 10,000. At 40,000 steps, the profit effect is statistically positive ($\Delta\pi = 55.358$, 95% CI [4.859, 105.857], positive in 85% of seeds), but the mean quantity change at that budget is small, unlike the 5,000 and 10,000-step rows where quantity moves substantially. This rules out the simplest explanation, that DQN is exploiting a fixed quantity-margin loophole from the start; the effect is not present early and does not have a single stable channel behind it at every budget. The more defensible interpretation is that the price cap becomes a focal constraint that some DQN runs learn to use only after sufficient training, rather than a channel visible from the beginning. A short trace is therefore insufficient evidence for this result; the training-budget ladder is the validation that makes the interpretation credible.

7.6 Secondary Pricing Interventions

Price cap is the only mechanism in this world that produces a consistent cross-learner pattern. The remaining four are correctly implemented and pass the same unit and parity checks as price cap, but their effects are smaller, more parameter-sensitive, or less consistent across learners. At the currently tested parameters, no single mechanism-level claim is as well supported for these interventions as it is for the price cap.

The high-price tax subtracts a quantity-weighted penalty from reward above a price threshold. Its effect is small in every direction and every learner: Q-learning's exploitability falls modestly, DQN is essentially unaffected, and the centralized critic moves in the wrong direction on price and welfare. At the tested tax rate, the penalty is too small, or too easy to avoid, to reliably change pricing behavior.

Random audits apply a stochastic penalty when average price is high. The result here is not only small but directionally inconsistent: for Q-learning, exploitability rises under audits rather than

Mind	Price Δ	Profit Δ	Exploitability Δ
Q-learning	−0.078	+2.2	−7.6
DQN	+0.006	+0.6	+1.2
PPO	−0.023	−3.6	−5.2
Independent-DQN	+0.001	+0.2	+2.5
Centralized critic	+0.088	+4.2	−1.2

Table 10: High-price tax versus no regulation, change by learner.

falling, and for the centralized critic, profit rises sharply while exploitability falls only partially. An enforcement mechanism with opposite signs across learners does not provide a robust guardrail at these parameters.

Mind	Price Δ	Profit Δ	Exploitability Δ
Q-learning	+0.114	+11.1	+32.0
DQN	+0.035	+4.3	−0.5
PPO	−0.006	−1.0	−2.7
Independent-DQN	+0.002	+0.2	+1.7
Centralized critic	+0.125	+49.1	−21.7

Table 11: Random audit versus no regulation, change by learner.

The direct anti-collusion penalty punishes prices that are both high and close together, the most explicit mechanism in the suite and, on paper, the one closest to actual antitrust enforcement logic. Its observed effect is limited: four of the five learners show negligible or worsened exploitability, and only PPO shows a mild improvement across price, profit, and exploitability together. A rule aimed directly at the behavior of interest is therefore not automatically more effective than the blunter price cap in this setting.

Mind	Price Δ	Profit Δ	Exploitability Δ
Q-learning	−0.048	−2.1	+31.9
DQN	+0.005	+0.5	0.0
PPO	−0.067	−7.4	−5.5
Independent-DQN	+0.001	+0.2	0.0
Centralized critic	−0.013	+0.1	+0.5

Table 12: Anti-collusion penalty versus no regulation, change by learner.

Demand shocks perturb market size with a lognormal multiplier each round, testing robustness to environmental volatility rather than acting as a policy lever. Results are small and mixed in direction across learners, with the centralized critic the only one meaningfully affected, and negatively. This mechanism is best read as a stress test the platform passes mechanically, not as a candidate institutional finding.

These outcomes should be interpreted as economic-activation results rather than implementation defects. All four mechanisms pass the same mechanics and parity tests as price cap. At the parameters currently tested, only price cap creates pressure strong enough to produce a consistent, cross-learner, interpretable signal. Table 14 summarizes the distinction.

Mind	Price Δ	Profit Δ	Welfare Δ	Exploitability Δ
Q-learning	+0.014	+4.0	+0.3	+11.4
DQN	−0.002	+1.3	+1.8	+1.3
PPO	−0.010	−0.3	+1.4	−6.0
Independent-DQN	+0.001	+0.6	+1.2	+1.8
Centralized critic	−0.050	−9.2	−12.0	+2.7

Table 13: Demand shock versus no regulation, change by learner.

Mechanism	Implementation	Interpretability
Price cap	Passes all checks	Strong, consistent, cross-learner
High-price tax	Passes all checks	Weak at tested rate
Random audit	Passes all checks	Mixed, sign-inconsistent
Anti-collusion penalty	Passes all checks	Limited binding, parameter-sensitive
Demand shock	Passes all checks	Robustness check, not a policy result

Table 14: Mechanism status, implementation versus interpretability, Pricing Arena.

7.7 Relation to the Main Question

Pricing Arena answers the main question through metric disagreement and training-horizon dependence. Price caps lower exploitability for every learner tested, but their effect on profit and collusion is learner-dependent, with the DQN effect emerging only after sufficient training. The remaining four mechanisms are correctly implemented, but at current parameters their effects are too small or inconsistent to support a general claim beyond the reported values. Whether the price-cap result and the secondary-mechanism patterns persist as the number of firms increases beyond two is addressed in the cross-world synthesis section.

8 Public Goods

8.1 World

Public Goods is a repeated common-pool resource and public-goods environment. In each round, agents choose between contributing to a shared pool, extracting from it, or remaining inactive. Contribution is individually costly but supports the shared state; extraction pays a private reward while depleting the pool. When requested extraction exceeds the available stock, extraction is rationed proportionally rather than allowed to run negative, and the pool regenerates deterministically each round. This makes the resource-state channel directly auditable.

The implemented institutions are no intervention, a penalty schedule aimed at extraction, contribution matching, reputation-based reward bonuses, an information-restriction variant that coarsens what agents observe about the pool, and a generic tax-and-redistribution schedule. The world records pool stock, sustainability, total contribution, total extraction, contribution and extraction rates, collapse-rate diagnostics, welfare, inequality, penalty totals, matched contribution, reputation bonuses, and tax revenue.

8.2 Known-Answer Validation

The benchmark obligations for Public Goods are bracketed rather than pointed: the implementation must distinguish free-riding from cooperative provision and must report welfare separately from the underlying resource state. The known-answer runner checks those regimes independently of any learned policy, at group sizes of two, four, and eight agents. All three checks pass. This does not validate that a learned population converges near the social optimum. It validates that, if a learned population moved toward that regime, the code would register the difference from free-riding correctly. The rest of this section therefore treats welfare movement as evidence only when it is accompanied by the relevant state variables.

8.3 Q-Learning Full Results

Table 15 shows the full $n = 20$ Q-learning run across all six institutions. The baseline confirms the intended commons pressure directly: contribution is low, extraction is high relative to it, and the collapse-rate diagnostic is above 0.9. Contribution matching moves furthest from that baseline on the state variables, raising sustainability, contribution, and welfare together while lowering collapse. Reputation raises welfare far more sharply than any other institution, but its effect on sustainability and contribution is much smaller than its effect on reward. The tax schedule produces little movement in the state variables while generating positive tax revenue, which is precisely the pattern this world's reporting discipline is intended to detect.

Institution	Sustainability	Contribution	Extraction	Welfare	Collapse
None	0.089	0.142	1.783	1.815	0.912
Penalty	0.088	0.126	1.762	1.785	0.919
Contribution matching	0.105	0.279	2.088	2.081	0.839
Reputation	0.094	0.222	1.885	9.575	0.873
Information restriction	0.088	0.125	1.760	1.799	0.920
Tax schedule	0.089	0.134	1.773	1.808	0.916

Table 15: Public Goods, Q-learning full run across all institutions.

Two institutions are particularly important for interpretation. The penalty schedule, although aimed directly at extraction, produces a slightly weaker outcome than no intervention at the tested parameters: welfare falls by 0.030 and collapse rises by 0.007 relative to baseline. This is not an implementation defect; the penalty is confirmed to bind, but it does not improve the commons under this configuration. Information restriction is close to baseline in every column and marginally weaker in most of them, so it does not support a positive institutional claim in the current run.

Reputation illustrates why this world reports welfare separately from the resource state. Its welfare value, 9.575, is more than five times the baseline welfare, but sustainability moves only from 0.089 to 0.094, and contribution moves from 0.142 to 0.222. The reputation bonus mean, 7.684, accounts for most of the welfare increase as an institutional reward, not as a consequence of substantially lower extraction or higher contribution. Tax schedule is a stricter version of the same diagnostic: tax revenue rises by 0.182, while sustainability, contribution, and welfare are approximately unchanged. Welfare, tax revenue, contribution, extraction, and sustainability therefore have to be interpreted as separate quantities.

8.4 Cross-Mind Results

Table 16 shows the no-institution baseline across the full learner suite. The main split is between learners that retain some contribution and learners that nearly eliminate it. PPO contributes 0.000, and centralized critic contributes 0.004, with collapse at or near 1.0 for both. DQN and independent-DQN sit between Q-learning and the two near-zero contributors, closer to Q-learning's level but still below it. In this world, additional function approximation or centralized training does not imply more contribution.

Mind	Sustainability	Contribution	Extraction	Welfare	Collapse
Q-learning	0.089	0.142	1.783	1.815	0.912
DQN	0.085	0.071	1.692	1.748	0.954
PPO	0.080	0.000	1.600	1.680	1.000
Independent-DQN	0.085	0.072	1.693	1.749	0.952
Centralized critic	0.080	0.004	1.606	1.684	0.997

Table 16: Public Goods, no-institution baseline, full learner suite.

Table 17 shows contribution matching for the same five learners. The institution changes behavior for Q-learning, DQN, and especially independent-DQN, whose contribution rises from 0.072 in the baseline to 0.413 under matching. It produces little response for PPO or the centralized critic. PPO's contribution remains 0.000, and centralized critic's contribution rises only to 0.007. The same rule, under the same environmental configuration, therefore changes the contribution channel for some learners and leaves it nearly inactive for others.

Mind	Sustainability	Contribution	Welfare	Collapse
Q-learning	0.105	0.279	2.081	0.839
DQN	0.096	0.170	1.923	0.904
PPO	0.080	0.000	1.680	1.000
Independent-DQN	0.121	0.413	2.269	0.805
Centralized critic	0.081	0.007	1.690	0.995

Table 17: Public Goods, contribution matching, full learner suite.

Reputation, shown in Table 18, makes the reward-versus-state distinction sharper when every learner is compared side by side. Every learner's welfare rises above its own baseline. For DQN, welfare rises from 1.748 to 12.444, higher than Q-learning's reputation welfare of 9.575. For PPO, welfare rises from 1.680 to 5.183 while contribution remains 0.000. A reputation-driven welfare increase is therefore not equivalent to improvement in the underlying commons state.

PPO's row is the clearest metric-decomposition case in this world: welfare more than triples relative to its own baseline through the reputation bonus, while contribution remains zero. If welfare were reported alone, PPO under reputation would appear successful. The contribution and sustainability columns show that this is a reward-accounting effect rather than a resolved commons problem.

8.5 Group-Size Status

Whether this pattern is a free-riding mechanism in the group-size sense, or a fixed-configuration outcome at the group size tested so far, remains open. A sweep over $n_{agents} \in \{2, 4, 8, 16\}$ was

Mind	Sustainability	Contribution	Welfare	Reputation Bonus
Q-learning	0.094	0.222	9.575	7.684
DQN	0.085	0.072	12.444	10.696
PPO	0.080	0.000	5.183	3.503
Independent-DQN	0.085	0.073	10.973	9.224
Centralized critic	0.080	0.006	3.976	2.290

Table 18: Public Goods, reputation system, full learner suite.

created to test this directly, but completed, validated summary outputs for that sweep are not available locally at the time of writing. Local split directories exist for each group size without validated summary files behind them. The claims in this section are therefore limited to the group size actually tested and validated here, and claims about how free-riding scales with group size are deferred until that sweep produces validated output.

8.6 Relation to the Main Question

Public Goods shows a commons failure that welfare alone can misdescribe. The baseline confirms the intended free-riding pressure directly, and contribution matching is the institution that most clearly moves the underlying resource state for three of the five learners tested. Reputation and tax schedule illustrate the opposite pattern at different scales: reputation produces a large welfare increase on top of a much smaller state change, while tax schedule produces almost no state change behind its revenue number.

The architecture split is substantial at the group size currently validated. PPO and centralized critic contribute at or near zero in the baseline and remain there under most institutions, including contribution matching. Q-learning, DQN, and especially independent-DQN respond to the same matching institution with increases in contribution and sustainability. Public Goods therefore supports the paper's central claim in a different form from Pricing Arena or Auction House: the guarantee that an incentive institution changes behavior is not architecture-invariant, and welfare cannot be read without the state variables the institution is meant to protect. Whether this split persists, sharpens, or narrows as group size changes is the open question the pending sweep is meant to answer.

9 Resource Island

9.1 World and Benchmark

Resource Island is a bounded-grid gather-and-trade economy. Agents move across the grid, gather food and wood, maintain an energy stock that is drawn down each round, hold separate food and wood inventories, and can starve if energy runs out. Available actions are staying in place, moving in one of four directions, gathering from the current cell, and offering food for wood or wood for food in trade. The world implements property rights, redistribution, trade price controls, and a reputation system as swappable institutions on top of this base economy. It tests the shared `World/Agent/Institution` interface outside a two-firm pricing game, in a setting with spatial structure, inventory, and multilateral exchange rather than a single simultaneous-move market.

Unlike Pricing Arena or Auction House, Resource Island has no closed-form external benchmark. Classical theory supplies qualitative expectations: property rights should matter when resources are contested, trade should raise welfare when agents have heterogeneous needs or access, reputation should support repeated exchange, and price controls can prevent unequal exchange while also blocking mutually beneficial trades. The benchmark obligation is therefore activation. An institution is interpreted only if the run creates the behavior the institution is meant to govern: contested resource access for property rights, unequal offers for trade controls, and repeated exchange for reputation.

The reported configuration is built around that obligation. It uses contested resource layouts, heterogeneous resource pressure, unequal trade offers, and direct counters for trade attempts, successful trades, inventory-blocked trades, institution-blocked trades, property claims, property violations, and property opportunities. These diagnostics make the Resource Island evidence activation-conditioned: welfare and survival are interpreted only together with the counters showing whether the relevant institutional channel was exercised.

9.2 Baseline

Table 19 shows the no-institution baseline under Q-learning, the reference point every institution result below is measured against. Trade attempts occur roughly twice as often as successful trades, survival is high, and resource sustainability sits below one, meaning agents draw the resource stock down through ordinary activity rather than leaving it untouched.

Metric	Value
Survival rate	0.9439
Welfare	1.1065
Successful trades	5.3251
Trade attempts	10.5507
Specialization index	0.3895
Inequality over time	0.1558
Resource sustainability	0.7638

Table 19: Resource Island, no-institution baseline, Q-learning.

9.3 Institution Results

Property rights create a claim on a cell after its first successful gather, and can then block or penalize a later gather attempt by a non-owner. This rule is measurably exercised: Table 20 reports nearly 28 property-opportunity events per run under Q-learning, alongside small positive changes in survival, welfare, and trade. Violations remain low relative to opportunities, so the result should not be interpreted as evidence that property rights resolve conflict in general. The supported claim is that the mechanism is observable and mildly beneficial in this configuration.

Trade price controls block any trade whose exchange ratio exceeds a set maximum. Table 21 shows that the control is active: successful trade falls to zero, and welfare, survival, and sustainability fall with it. The institutional channel is therefore exercised, but the observed effect is restrictive rather than welfare-improving. A rule designed to prevent unequal exchange can eliminate exchange altogether rather than making it more equitable, and that is the pattern observed in this configuration.

Reputation gives agents a reward bonus tied to a history of successful trades. Table 22 shows welfare and trade both rising under this institution. This separates the result from a purely

Metric	Value
Survival Δ	+0.0077
Welfare Δ	+0.0474
Trade count Δ	+0.4861
Property claims	1.8695
Property violations	0.0468
Property opportunities	27.8249

Table 20: Property rights versus no institution, Q-learning.

Metric	Value
Survival Δ	−0.0249
Welfare Δ	−0.1558
Trade count Δ	−5.3251
Successful trades (level)	0.0000
Institution-blocked trades	4.7862
Resource sustainability Δ	−0.0300

Table 21: Trade price controls versus no institution, Q-learning.

accounting-based reward effect: trade volume also changes, so at least part of the welfare gain is associated with additional exchange.

Metric	Value
Survival Δ	−0.0138
Welfare Δ	+0.2412
Trade count Δ	+1.3414
Trade attempt count Δ	+0.5410

Table 22: Reputation system versus no institution, Q-learning.

Redistribution is not part of the main activated Resource Island institution comparison reported here. It remains implemented in the world, but the current thesis-facing Resource Island evidence focuses on property rights, trade price controls, and reputation because those are the institutions with directly validated activation diagnostics in the reported table.

9.4 Cross-Mind Results

Table 23 shows the no-institution baseline across the full learner suite. The split is substantial: Q-learning, PPO, and the centralized critic all sustain meaningful trade, while DQN and independent-DQN trade much less, by roughly an order of magnitude on trade count relative to PPO. DQN and independent-DQN also show the highest resource sustainability in the table, but this should be read as reduced economic activity rather than as efficient resource governance.

The same environment supports high trade volume for three of the five learners, which is why the low DQN and independent-DQN trade counts are not interpreted as evidence that the world lacks trade opportunities. They are evidence that, at the tested budget, those learners use the exchange channel much less than Q-learning, PPO, or the centralized critic. This is a different architecture

Mind	Survival	Welfare	Trades	Attempts	Specialization	Sustain.
Q-learning	0.9439	1.1065	5.3251	10.5507	0.3895	0.7638
DQN	0.9102	1.0614	0.6398	1.0709	0.1790	0.9529
PPO	0.9150	1.2537	8.8919	9.8380	0.4052	0.8789
Independent-DQN	0.9064	1.0630	1.7300	2.1310	0.1524	0.9639
Centralized critic	0.9150	1.2484	8.6973	10.8525	0.4073	0.8920

Table 23: Resource Island, no-institution condition, full learner suite.

effect from the one observed in Pricing Arena. In Pricing Arena, DQN eventually learns a profit-preserving response to a price constraint. In Resource Island, DQN and independent-DQN underuse a productive channel that the other learners demonstrably use. The failure mode is non-discovery or weak reinforcement of a productive market opportunity rather than exploitation of a regulatory guardrail.

Table 24 shows that this split is not confined to the baseline; it carries through to institutional activation. Property opportunities under property rights, and trade volume under reputation, both show the same DQN-and-independent-DQN-low pattern as the baseline trade numbers.

Mind	Property Opportunities	Trade (Reputation)	Welfare (Reputation)
Q-learning	27.8249	6.6665	1.3477
DQN	8.2279	2.4212	1.1686
PPO	27.1268	13.9870	1.8244
Independent-DQN	6.5695	2.2882	1.1541
Centralized critic	24.9577	7.7649	1.4858

Table 24: Institution activation by learner, property rights and reputation.

Reputation amplifies an exchange channel that already has to exist for the reputation bonus to have behavioral content. For DQN and independent-DQN, the baseline trade channel is limited, so the amplification is limited as well. Trade price controls show the opposite pattern: successful trade falls to zero for every learner under this institution, so the formal rule binds universally, but the amount of blocked-trade pressure differs sharply by learner, from 4.79 for Q-learning to 0.04 for PPO. The practical force of an identical rule therefore depends on how much a given learner attempts to trade in the first place.

9.5 Relation to the Main Question

The DQN and independent-DQN trade result above is reported at a single training budget. Pricing Arena's price-cap result showed that a single-budget reading of a learning-architecture effect can be misleading: a result that looked negative at 1,000 steps became positive and statistically significant at the full 40,000-step budget. The same caution applies here. Learning to trade in Resource Island requires a sequence of actions: moving to resources, gathering, constructing an offer, and finding a compatible counterparty. DQN's off-policy, replay-buffer exploration could plausibly discover and reinforce that sequence more slowly than PPO's on-policy exploration, without that implying that DQN cannot learn it at all.

Until a budget ladder analogous to Table 9 is run for this world, the claim is limited to the training budget actually tested. At that budget, trade price controls bind and eliminate exchange, reputation amplifies trade where a learner already engages in it, and property rights are measurable through opportunity counters. The learner split is also activation-conditioned: Q-learning, PPO, and the centralized critic sustain meaningful trade economies, while DQN and independent-DQN show low trade activation relative to those learners. Whether that gap narrows, holds, or reverses with more training remains open.

10 Cross-World Synthesis

The emerging thesis is not that one institution always works, or that the agent classes form a universal intelligence ranking. The stronger claim is more specific: institutional robustness must be evaluated jointly with the learning architecture, the training horizon, and activation diagnostics. Tabular value learning, function approximation, on-policy policy gradients, decorrelated independent learners, and centralized critics violate different assumptions of the classical benchmark story. The comparisons are therefore best read as a learner-architecture stress test, not as a scalar capability ranking.

The Pricing Arena result shows architecture-sensitive institutional effects: price caps reduce exploitability broadly, but DQN's profit response flips sign with training budget and is positive in the full paired audit. Resource Island shows why activation diagnostics are necessary: institution effects are interpretable only in runs where agents attempt the behavior the institution is meant to govern. Auction House adds theory-anchored bidding benchmarks [Vickrey, 1961, Myerson, 1981]. Public Goods adds a sustainability and commons-collapse channel. Labor Market adds a matching-stability and truthfulness channel.

The five-world synthesis is now built from full n=20 learner-suite outputs for Pricing Arena, Resource Island, Auction House, Public Goods, and Labor Market. The protocol-comparability report passes for all five worlds, and the publication inference table includes all worlds under the same confirmatory bookkeeping. The claim-audit suite adds a second pass over those outputs: paired seed effects, benchmark gaps, metric-conflict warnings, activation confirmations, and trace/full-run sign agreement where paired traces exist. In its current run, the audit produces 1,025 paired or benchmark rows and 31 metric-conflict or activation rows. This audit layer prevents the cross-world synthesis from becoming a table of averages without claim boundaries.

The main synthesis claim is qualitative but concrete. Learning architecture changes institutional outcomes in every world, but not through one universal direction. In Pricing Arena, DQN's price-cap profit increase is a long-training, seed-heterogeneous effect rather than a short-run channel. In Resource Island, PPO and centralized-critic activate trade strongly while DQN-family learners trade much less [Mnih et al., 2015, Schulman et al., 2017]. In Auction House, the relevant question is distance from truthful, shaded, or reserve-price benchmarks rather than raw reward [Vickrey, 1961, Myerson, 1981, Milgrom, 2004]. In Public Goods, the relevant state variables are contribution, extraction, and sustainability, while some tax effects are accounting effects rather than state changes. In Labor Market, deferred acceptance produces high match rates and mostly stable outcomes, so future manipulation claims must target the right side of the market or a different matching institution.

11 Limitations

Several limitations are deliberate. The environments are small and stylized. They are designed for auditability rather than realism. The deep-RL agents are structured-observation MLP agents rather than large-scale policies. The Resource Island trade protocol is still simple, and the current pressure-tested configuration is an activation test rather than a realistic market. Auction House currently has benchmarked sealed-bid full runs and smoke-tested clock and information variants. Public Goods is still a stylized common-pool model rather than an empirical commons. Labor Market currently uses worker-side learning under deferred acceptance, so it tests one asymmetric matching channel rather than matching-market design generally.

The most important limitation is interpretive. A complete CSV does not imply a validated economic result. Every institution must be checked for activation. Every learner-architecture comparison must be checked for implementation artifacts. The independent-DQN alias issue is a concrete example: the table was statistically valid as an output file, but not valid as evidence for a distinct mind until the implementation was corrected and rerun.

The Pricing Arena price-cap audit adds a second interpretive warning. A mechanism trace can reveal the local channel but cannot replace a replicated training-budget audit. The 1000-step trace correctly showed early profit losses under the cap; the 40,000-step table correctly showed a positive DQN cap-minus-none profit delta. The result is reportable only because the contradiction was resolved through paired seed deltas and a budget ladder.

12 Reproducibility

The repository records validation status in `ROADMAP_STATUS.md`. Core outputs are stored under `outputs/`. Major runners include:

- `run_multiseed.py` for Pricing Arena replicated runs;
- `run_exploitability.py` for frozen-policy adversarial evaluation;
- `run_phase3_full.py` and `validate_phase3_full.py` for full Phase 3 Pricing Arena checks;
- `run_known_answer_sanity_checks.py` for theorem/bracketing checks before learned-policy interpretation;
- `run_mechanism_traces.py` for per-step mechanism traces and decomposition tables across worlds;
- `run_pricing_quantity_margin_audit.py` for the Pricing Arena price-cap budget-ladder contradiction audit;
- `run_claim_audit_suite.py` for cross-world paired-effect, benchmark-gap, metric-conflict, activation, and trace-alignment audits;
- `run_resource_island_smoke.py` for Resource Island smoke, validation, and full runs;
- `run_auction_house_smoke.py` for Auction House runs;
- `run_public_goods_smoke.py` for Public Goods runs;
- `run_labor_market_smoke.py` for Labor Market runs;
- `build_world_mind_comparison.py` and `build_cross_world_synthesis.py` for ladder comparisons and cross-world synthesis.

The test suite currently includes unit, integration, and output-validation coverage for the core abstractions, Pricing Arena, Resource Island, Auction House, Public Goods, Labor Market, and deep-RL/MARL minds.

12.1 Primary Output Artifacts

The main numerical claims in the paper are backed by the following generated artifacts:

- `outputs/known_answer_sanity_checks_current/summary.csv`;
- `outputs/mechanism_traces_local_full/mechanism_decomposition.csv`;
- `outputs/phase3_full/mind_comparison.csv`;
- `outputs/pricing_quantity_margin_audit/dqn_budget_ladder.csv`;
- `outputs/pricing_quantity_margin_audit/existing_full_seed_deltas.csv`;
- `outputs/resource_island_v1_full/summary_aggregate.csv`;
- `outputs/resource_island_v1_strict_radius_full/summary_aggregate.csv`;
- `outputs/resource_island_v1_phase3_full/mind_comparison.csv`;
- `outputs/auction_house_phase3_full/mind_comparison.csv`;
- `outputs/public_goods_phase3_full/mind_comparison.csv`;
- `outputs/labor_market_phase3_full/mind_comparison.csv`;
- `outputs/cross_world_synthesis/synthesis_table.csv`;
- `outputs/claim_audit_suite/claim_audit_report.md`.

13 Conclusion

This draft reports a first version of an artificial-economies platform for testing institutional robustness under learning agents. The mature Pricing Arena result already shows why the question matters: exploitability and collusion can move differently under the same institution, and a headline effect can depend on training budget rather than appearing in short traces. Resource Island, Auction House, Public Goods, and Labor Market extend the project beyond pricing games. Their role is not merely to add more checkboxes, but to test whether the patterns found in Pricing Arena survive when the economic structure changes and when the relevant institutional channel is actually activated.

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